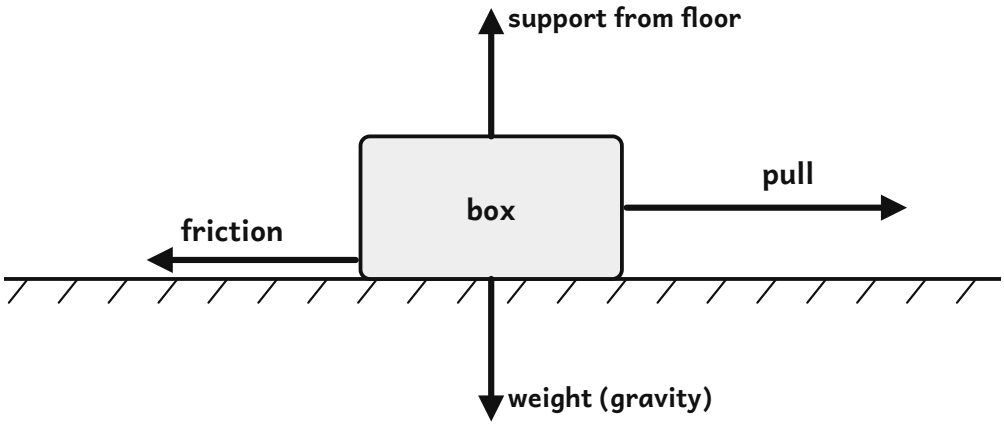
PHYSICAL AND CHEMICAL SCIENCE

Forces and Motion

Pushes and pulls, measuring forces, gravity, friction, weight and motion



In this pack

- **Learn it:** Pushes and pulls ★☆☆
- **Fact check:** Push or pull? ★☆☆
- **Think it through:** Forces all around ★★☆
- **Investigate:** How far does the car roll? ★★☆
- **Real-world case study:** The rickshaw van ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science: Stage 3 Unit 6 Forces and movement (6.1–6.5); Stage 2 Unit 3 Changing materials (3.1–3.3).

This pack belongs to: _____ **Class:** _____

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Pushes and pulls

Name: Class: Date:

Learning goals

- Explain that pushes and pulls are forces, and that forces can be measured.
- Show how forces start, stop and change the shape of things.
- Describe gravity, friction and weight.

Kick a ball, open a door, pull a rickshaw van, squash some clay. Every time, you use a **force**. A force is a **push** or a **pull**.


kick: push


push


pull the oars


pull the string

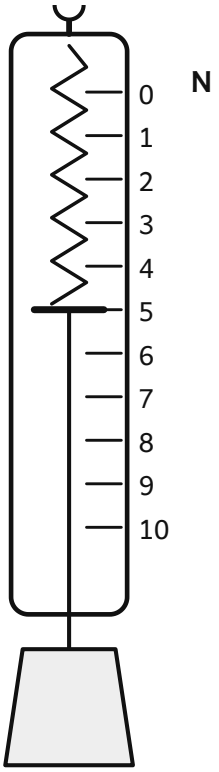

hit: push

What forces do

- make things **start** moving (push a swing);
- make things **stop** (catch a ball);
- make things go **faster** or **slower**;
- change the **direction** things move (hit a ball with a bat);
- change the **shape** of things (squash, stretch, bend or twist clay).

Measuring forces

We can measure how big a force is with a **forcemeter**. It has a spring inside. The harder you pull, the more the spring stretches. Forces are measured in **newtons** (N).



A forcemeter.

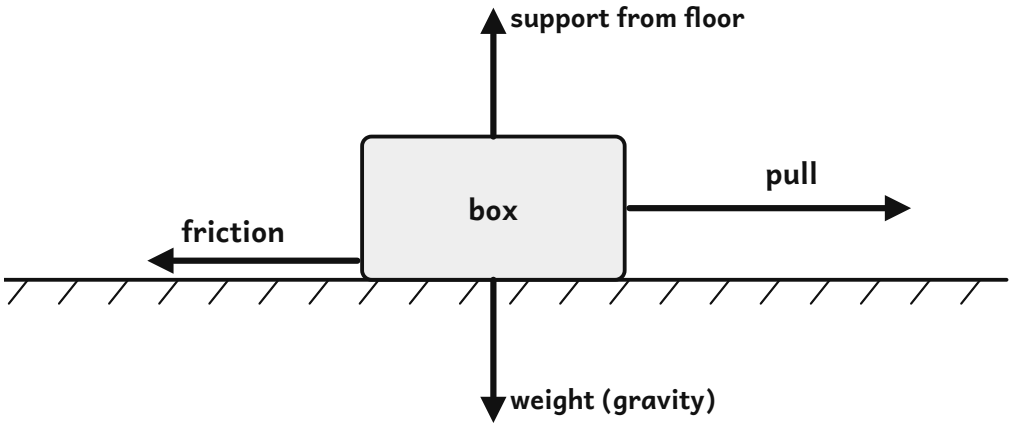
A heavy bag needs a **bigger** force to lift or move than a light bag. The **heavier** something is, the more force you need to move it.

Gravity

Gravity is a force that pulls everything **down** towards the Earth. That is why a mango falls from the tree and a ball comes back down. Your **weight** is how hard gravity pulls on you.

Friction

Friction is a force that happens when two things rub together. It **slows things down**. A ball rolls far on a smooth floor, but stops quickly on grass or sand. Friction helps us: our shoes grip the ground so we do not slip, and bicycle brakes stop the wheels.



Forces on a box being pulled.

Motion

Motion means moving: changing position over time. A car that goes further in the same time is moving **faster**.

Key words

force	a push or a pull
gravity	pulls things down to the Earth
friction	slows things when they rub
weight	how hard gravity pulls on something
forcemeter	measures forces in newtons

FACT CHECK · C2 · PUSHES AND PULLS

Push or pull?

★☆☆
Grade 2

Name: _____ **Class:** _____ **Date:** _____

Learning goals

- Sort pushes and pulls.
- Name forces.

1 Push or pull? [6]

SORT THESE

pushing a pram **kicking a ball** **opening a drawer** **pulling a rope**
pressing a button **flying a kite**

Push	Pull

2 What does the force do? Draw a line. [4]

Draw a line to match each pair, or write the letter.

squashing clay ●	● **A** changes its shape
catching a ball ●	● **B** starts it moving
pushing a swing ●	● **C** stops it
hitting a ball with a bat ●	● **D** changes its direction

3 Fill in the gaps. [3]

WORD BANK

gravity **friction** **pull** **light**

A mango falls down because of _____. A ball stops on grass because of _____. A force is a push or a _____.

Forces all around

★★☆
Grade 2

Name: Class: Date:

Learning goals

- Explain friction and gravity.
- Compare forces.

1 **Why is it easier to slide on a wet, smooth floor than on a rough mat?** [2]

2 **Which needs a bigger force to push? Why?** [2]



empty cart



cart full of heavy boxes

3 **Draw an arrow to show which way gravity pulls a falling mango.** [1]

Falling mango

INVESTIGATE · C2 · PUSHES AND PULLS

How far does the car roll?

★★☆
Grade 2

Name: Class: Date:

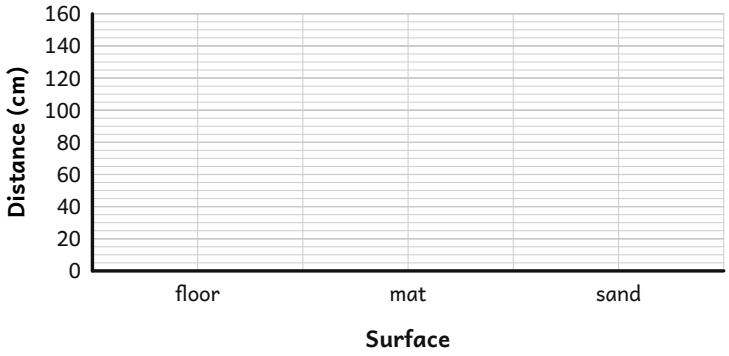
Learning goals

- Do a fair test.
- Measure distance.

Class 2B rolled a toy car down a ramp onto three surfaces and measured how far it went.

1 **Draw a bar chart.** [3]

Surface	smooth floor	mat	sand
Distance (cm)	150	80	20



2 **On which surface did the car go furthest?** [1]

.....

3 **Which surface had the most friction?** [1]

.....

4 **Why did they use the same ramp and car each time?** [1]

.....

REAL-WORLD CASE STUDY · C2 · PUSHES AND PULLS

The rickshaw van

★★★ Grade 2

Name: **Class:** **Date:**

Learning goals

- Find forces in real life.

Science | Work | Maths

Kamal drives a **rickshaw van** that carries vegetables to the market. He **pushes** the pedals with his feet to make it go. When the van is full of potatoes it is very **heavy**, and he must push much harder.

When he wants to stop, he squeezes the **brakes**. The brakes rub on the wheels and **friction** slows the van down. On a wet road the van slides more, so Kamal goes slowly.

A story about forces at work.

1 **What force does Kamal use to make the van go?** [1]

2 **Why is it harder to pedal when the van is full?** [2]

3 **How do the brakes stop the van?** [2]

Show what you know

★★★
Grade 2

Name: **Class:** **Date:**

Learning goals

- Use force words.

1 Find the words.

O	T	E	O	U	O	G	E
U	O	M	T	L	Y	F	H
B	S	L	O	W	V	L	I
M	G	R	A	V	I	T	Y
E	F	D	F	P	U	L	L
C	A	I	F	O	R	C	E
I	S	S	P	U	S	H	C
E	T	B	T	E	O	H	O

fast force gravity pull push slow

2 Which force pulls things down? [1]

◯ friction ◯ gravity

◯ a push

3 Squashing clay changes its... [1]

◯ colour ◯ shape

◯ weight

4 Challenge. Draw yourself pushing something and pulling something. Label each. [2]

push

pull

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort pushes and pulls	○	○	○
say what forces can do	○	○	○
explain gravity	○	○	○
explain friction	○	○	○
test how far a car rolls	○	○	○